Wenhao Zhang

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RESEARCH INTERESTS

Vision foundation models; vision-language and multimodal representation learning; data-efficient training, fine-tuning, and evaluation of large-scale visual models.

EDUCATION

University of Virginia
Ph.D. in Data Science. Advisor: Prof. Sheng Li — Aug 2025 – May 2029 (Expected)

Xidian University
M.E. in Artificial Intelligence – GPA: 90.42/100 — Aug 2022 – June 2025

Xidian University
B.E. in Computer Science – GPA: 3.7/4.0 — Aug 2018 – June 2022

RESEARCH EXPERIENCE

University of Virginia - Graduate Research Assistant — Charlottesville, VA - Aug 2025 - Present

- Built CleanSlide, a public vision-language benchmark of 149K multiple-choice questions over 9,985 gigapixel images across 32 categories, with patient- and site-disjoint splits and published leakage and shortcut audits; encoder features released on Hugging Face (github.com/wenhaozhang0066/CleanSlide).
- Audited train-test contamination across the public vision-language benchmark ecosystem, documenting 92.3-100% case-level overlap and near-complete institutional overlap; showed that a parameter-free text retriever reproduces most of the reported accuracy without ever seeing the image (WACV 2027, Datasets and Benchmarks Track).
- Built a three-level LLM-based image description pipeline over 15 public gigapixel image datasets, producing Balance9k, a disentangled image-text pair dataset for vision-language pre-training.
- Fine-tuned DINOv3 vision transformers in PyTorch on Balance9k; showed that 9K curated samples match the downstream performance of million-scale fine-tuning, two orders of magnitude less data.
- Introduced Cell-Guided Attention, a region-guided attention module, together with domain-agnostic training, improving the model's generalization to unseen visual domains by 30% on average, and Text-Guided Pooling to suppress hallucination in LLM-generated captions.
- Maintain Awesome-Pathology-VLMs, a curated open-source index of pathology vision-language models, datasets, and benchmarks (github.com/wenhaozhang0066/Awesome-Pathology-VLMs).

Xidian University - Graduate Research Assistant — Xi'an, Shaanxi - Aug 2022 - June 2025

- Proposed PedMix, a cross-modal data augmentation strategy, and a Modality Feature Transfer module combining cross-attention with convolutional backbones, for visible-infrared person re-identification; improved mAP by 4.67% (Scientific Reports 2025).
- Designed a lightweight directional-aware network reaching state-of-the-art classification accuracy at significantly lower computational cost (Artificial Intelligence Review 2024).

University of Nebraska Medical Center - Intern in Biometrics — Omaha, NE - Dec 2023 - May 2024

- Built a patch-wise regression pipeline over 3D volumetric image data, replacing convolutional stages of a local U-Net with Swin Transformer blocks to predict subject age from local image features.

TECHNICAL SKILLS

Programming: Python, C++, Bash, LaTeX

Deep Learning & CV: PyTorch, Hugging Face, timm, NumPy, scikit-image; Vision Transformers (ViT, DINOv3, Swin), CNNs, U-Net, vision-language models, cross-attention, self-supervised and data-efficient fine-tuning

Tools: Git, Linux, Docker, Slurm, Weights & Biases

PUBLICATIONS

W. Zhang, Z. Zhou, S. Li. "Auditing Data Leakage in Whole-Slide Image Multimodal Benchmarks." Under review, WACV 2027 (Datasets and Benchmarks Track).

W. Zhang, Z. Zhou, S. Li, et al. "CleanSlide: A Leakage-Audited and Shortcut-Controlled Benchmark for Whole-Slide Vision-Language Models." In preparation, 2026.

W. Zhang, Z. Zhou, S. Li, et al. "Vision-Language Models for Computational Pathology: A Survey of Paradigms, Granularities, and Benchmarks." In preparation, 2026.

W. Zhang, Z. Zhou, S. Li. "LANTERN: Learning a Data-Efficient Pathology Foundation Model through Multimodal Disentangled Representations." In preparation, 2025.

Y. Guo, **W. Zhang**, L. Jiao, J. Du, S. Wang, S. Wang, L. Fang. "Visible-Infrared Person Re-Identification with Region-Based Augmentation and Cross-Modality Attention." Scientific Reports, 2025.

Y. Guo, W. Zhan, L. Jiao, **W. Zhang**, K. Chen, J. Jing, L. Fang. "Multi-Distribution Time-Series Prototype Learning for Crop Mapping with Sentinel-1 SAR Imagery." IEEE Transactions on Geoscience and Remote Sensing, 2025.

Y. Guo, **W. Zhang**, Y. Gao, L. Jiao, R. Qu, S. Wang, F. Liu, J. Du. "Revisiting Non-Learned Operators Based Deep Learning: A Lightweight Directional-Aware Network for Classification." Artificial Intelligence Review, 2024.

PATENTS

Y. Guo, **W. Zhang**, J. Du, W. Zhan. "Cross-Modal Pedestrian Re-Identification Method Based on Spatial Data Augmentation and Symmetric Cross-Modal Attention." CN Patent, Application No. 202310838299.7, 2023.

Y. Guo, J. Du, **W. Zhang**, Y. Gao. "Causal Inference-Based Semi-Supervised Image Classification Method." CN Patent, Application No. 202210896650.3, 2022.

Y. Guo, Y. Gao, J. Du, **W. Zhang**. "Image Classification Method Based on Shearlet Networks and Directional Attention Mechanism." CN Patent, Application No. 202211044682.7, 2022.

ACADEMIC SERVICE

Workshop Co-Organizer: 1st Workshop on Big Data for Computational Pathology and Multi-Omics: From Pixels to Biomarkers, IEEE BigData 2026, Phoenix, AZ.

Conference Reviewer: CVPR 2026, ACM MM 2026, ACL Rolling Review (ARR) 2026